Week 4 SQL Mini Project Report

1. Project Overview

This project focuses on applying SQL concepts to solve real-world retail business problems. The dataset used was a retail sales transactions dataset containing customer, product, and sales-related information. The project included data validation, business analysis, CASE WHEN statements, CTE queries, and sales trend analysis.

2. Dataset Information

Column	Description
Transaction ID	Unique transaction identifier
Date	Transaction date
Customer ID	Unique customer identifier
Gender	Customer gender
Age	Customer age
Product Category	Category of purchased product
Quantity	Quantity purchased
Price per Unit	Price of each unit
Total Amount	Total transaction amount

3. Data Validation and Cleaning

Initial validation checks were performed before analysis. The dataset was checked for:

- NULL values in important columns
- Duplicate transaction IDs
- Dataset structure and category values

The dataset contained no NULL values and no duplicate transaction IDs, indicating that the data was clean and ready for analysis.

4. SQL Concepts Used

- GROUP BY
- ORDER BY
- Aggregate Functions (SUM, AVG, COUNT)
- CASE WHEN
- CTE (Common Table Expression)

5. Business Insights

1 The Clothing category generated the highest revenue among all product categories.

2. A small group of customers contributed significantly higher spending.
3. Sales contribution between male and female customers was relatively balanced.
4. Adult customers generated the highest revenue compared to other age groups.
5. Some product categories performed above the overall average revenue.
6. Monthly sales analysis showed variations in revenue across different months.
7. Some product categories recorded higher quantities sold, indicating stronger customer demand.

6. Conclusion

This project demonstrated how SQL can be used to analyze retail transaction data and generate business insights. The analysis helped identify high-performing product categories, customer purchasing behavior, and revenue trends using SQL-based data analysis techniques.

7. Learning Outcome

Through this project, important SQL concepts such as aggregation, filtering, CASE WHEN statements, and CTEs were practiced in a real-world business scenario. The project also improved understanding of data validation, business analysis, and basic data storytelling.